

A Comprehensive Measure Of Well Being

Human Development Index Prediction System

Project	A Comprehensive Measure of Well Being
Team	AIML
Team Lead	Gandi Manasa Venkata Rama Lahari Sri
Team Size	4

Introduction

The Human Development Index (HDI) is a statistical composite index of life expectancy, education, and per capita income indicators, which are used to rank countries into four tiers Very High, High, Medium, and Low of human development. A country achieves a higher HDI when its population enjoys a longer lifespan, better educational attainment, and a higher Gross National Income (GNI PPP) per capita.

The HDI was developed to emphasize that people and their capabilities should be the primary measure of a country's development rather than economic growth alone. It provides a broader perspective on development by considering health, education, and living standards together. The index is widely used by governments, researchers, policymakers, and international organizations to evaluate progress, compare countries, and identify areas requiring improvement.

Key Features

- ❑ AI-Based HDI Prediction using the Random Forest Machine Learning algorithm.
- ❑ Real-Time Prediction based on Life Expectancy, Education, and GNI indicators.
- ❑ HDI Category Classification into Low, Medium, High, and Very High.
- ❑ Interactive Dashboard displaying prediction results with professional charts.
- ❑ Data Visualization using Bar Charts and Pie Charts for better analysis.
- ❑ User Authentication with secure Registration and Login functionality.
- ❑ Prediction History to store and view previous predictions for each user.
- ❑ AI Development Insights providing analysis and recommendations based on prediction results.
- ❑ Responsive User Interface with Light/Dark Mode support for a better user experience.
- ❑ Secure Database Management using SQLite for user and prediction data storage.

Usage Scenarios

Scenario 1: Predicting Very High Human Development

A user selects a country with high life expectancy, strong mean years of schooling, expected years of schooling, and a high GNI per capita. Based on these indicators, the model predicts a Very High HDI score, classifying the country among the most developed nations. This helps users understand how strong performance across multiple development dimensions contributes to overall human development.

Scenario 2: Identifying Development Gaps in Emerging Economies

A policymaker inputs mid-range values representing a developing nation, including moderate life expectancy, average educational attainment, and a moderate income level. The model returns a Medium HDI score, providing insights into areas where improvements in healthcare, education, or income generation could significantly enhance human development outcomes.

Scenario 3: Assessing Countries Requiring Development Intervention

A researcher evaluates a country with relatively low life expectancy, limited educational opportunities, and low GNI per capita. The model predicts a Low HDI score and highlights the country's developmental challenges. Such predictions can support governments and development organizations in prioritizing investments and policy interventions.

Team

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Skills Involved

- ☐ Python
- ☐ HTML5 ,CSS3,JavaScript
- ☐ Flask
- ☐ Bootstrap 5
- ☐ SQLite
- ☐ Machine Learning
- ☐ Pandas ,Numpy
- ☐ Data Visualization
- ☐ Git & GitHub,VS Code

System Requirements

Hardware Requirements

Processor: Intel Core i3 or above

RAM: Minimum 4 GB

Storage: Minimum 10 GB free space

Internet Connection for downloading packages and datasets

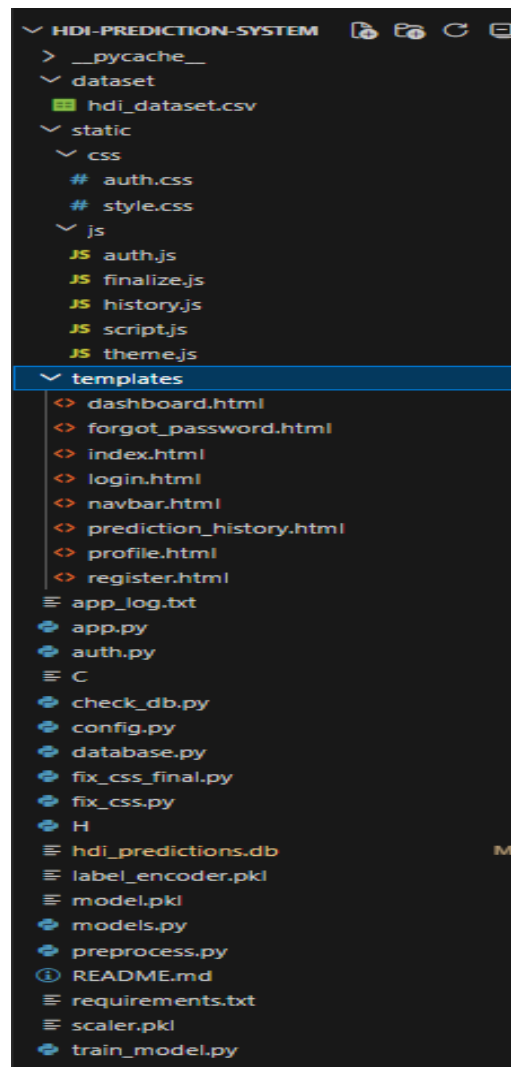
Software Requirements

Operating System: Windows / Linux / macOS

Python 3.x

Flask Framework

Project Folder Structure:



VS Code project explorer showing the HDI-prediction-system folder structure

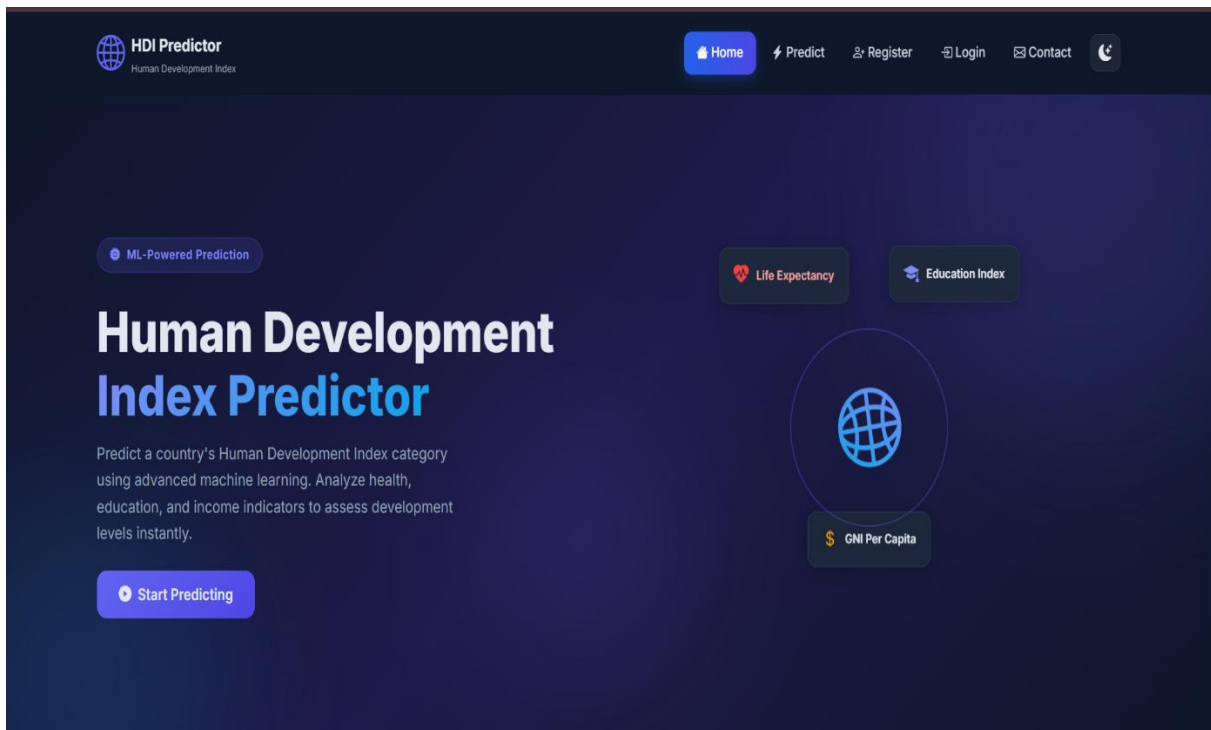
Running the Application Locally

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS CODE REFERENCE LOG SPELL CHECKER
PS C:\Users\manas\OneDrive\Documents\Desktop\HDI-prediction-system> python app.py
[OK] Model artifacts loaded successfully.
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.5.34.212:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
[OK] Model artifacts loaded successfully.
* Debugger is active!
* Debugger PIN: 416-925-482
```

Once running, the application is accessible in the browser at **127.0.0.1:5000**

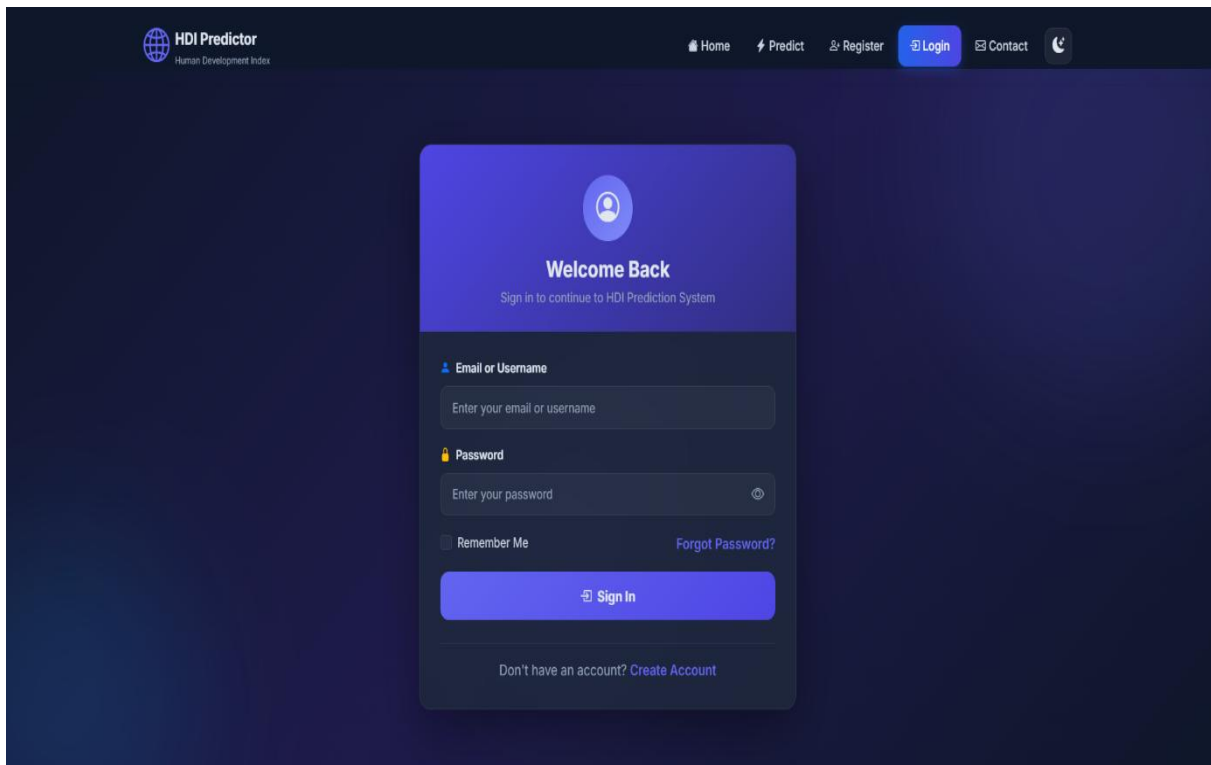
User Interface Screenshots

Home Page



Sign In / Sign Up

Users can sign in with an email/password or via Google Sign-In, with a matching Sign Up flow for new accounts.



"Welcome Back" sign-in modal on the HDI Prediction home page

Prediction page



HDI Predictor
Human Development Index

ML PREDICTION

[Home](#) [Predict](#) [Register](#) [Login](#) [Contact](#)

Predict HDI Category

Enter the development indicators below to get an instant ML-powered HDI prediction.

Input Indicators

📍 Country

India

Start typing to search countries. Country value does not affect prediction.

❤️ Life Expectancy

67

Range: 20 - 90 years

Average

📖 Mean Years of Schooling

11

Range: 0 - 20 years

Moderate

🎓 Expected Years of Schooling

12

Range: 0 - 25 years

Moderate

💰 GNI Per Capita

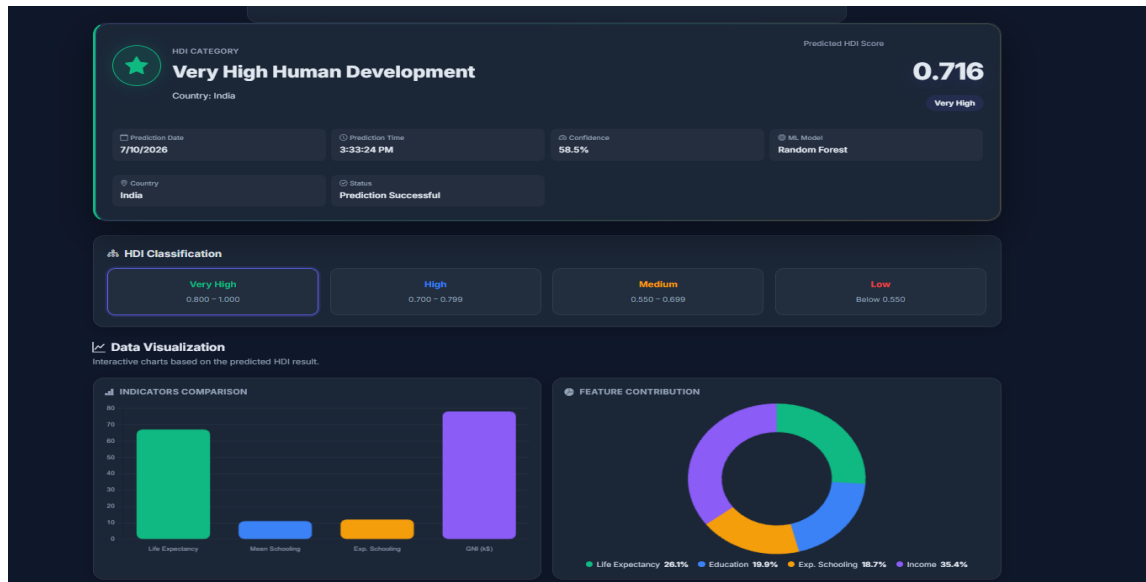
78000

Range: \$100 - \$100,000

High

Predict HDI Category

Output/Result



Known Issues

- Demo and GitHub links not yet added for mentor review
- No mentor assigned yet
- Notification system for reminders not yet implemented

Future Enhancements

- ☐ Integrate **Deep Learning models** to improve prediction accuracy.
- ☐ Support **real-time HDI data** through APIs from organizations such as the UN or World Bank.
- ☐ Add **advanced analytics dashboards** with more interactive visualizations and trend analysis.
- ☐ Implement **PDF and Excel report generation** for prediction results.
- ☐ Develop a **mobile application** for Android and iOS users.
- ☐ Add **country-wise HDI comparison** and historical trend analysis.
- ☐ Improve security by implementing **JWT/OAuth authentication** and role-based access control.
- ☐ Integrate an **AI chatbot** to provide personalized insights and recommendations based on prediction results.

Conclusion

The **HDI Predictor – Human Development Index Prediction System** is a Machine Learning-based web application that provides accurate and efficient prediction of a country's HDI category using key socio-economic indicators. By integrating the **Random Forest algorithm** with a **Flask-based web application**, the system delivers real-time predictions, interactive visualizations, user authentication, and prediction history. The project simplifies HDI analysis through a user-friendly interface and supports informed decision-making with AI-generated insights. Overall, it demonstrates the effective integration of **Machine Learning, Full-Stack Web Development, Database Management, and Data Visualization** to build a reliable and practical HDI prediction solution.